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Data:

1. [Reddit MBTI Dataset](#) (public): Reddit posts labeled with self-reported MBTI personality types.
2. [Emotion-Balanced Twitter Dataset](#) (public): short texts labeled with six emotions (sadness, joy, love, anger, fear, surprise).

Background and Motivation: Personality theory suggests that individuals differ in stable emotional tendencies. Rather than directly predicting personality from text, this project proposes first learning a robust model of emotional expression and then using it to measure how emotional language differs across personality types. This approach treats emotion as a measurable intermediate representation rather than a secondary outcome, allowing us to study how personality is reflected in affective language patterns.

Problem: This project will aim to explore whether emotional distributions differ systematically across personality categories and whether emotional features improve personality modeling. Broadly, this will involve training a language model to classify emotional content, applying it to Reddit posts from users with known MBTI types, aggregating emotional signals at the user level, and analyzing whether personality types exhibit distinct emotional language profiles.

Scope and Methods: We'll use methods covered in the course, including but not limited to transformer-based language models, transfer learning, multi-class classification, and representation analysis and clustering.

Concerns and Limitations: MBTI labels are self-reported and imperfect measures of personality. Further, emotional classifiers trained on one platform may not perfectly generalize to another. Our inference will be based on published text, which may not perfectly translate to an individual's "true" emotions. The emotion dataset is effectively anonymized, while the Reddit MBTI dataset is pseudonymized. No attempt will be made to identify users, and results will only be reported at the aggregate level.